

21. (1.75 points)

A company writes two lines of business: personal auto and personal property. An actuary simulated 1,000 loss scenarios of a company's personal property line of business using an Internal Risk Model. Given the following information:

- 1-in-100 TV@R of personal auto = \$4.5 million.
- 1-in-100 WTV@R of personal auto = \$5 million.
- The company has a total capital of \$10 million.
- All scenarios have an equal chance of occurring.
- The actuary calculated the transformed probability function ($F^*(x)$) using Wang transform.
- The results of the 13 worst scenarios sorted in descending order for personal property:

Simulation #	Loss	$F^*(x)$
709	6,933,000	1.000
330	5,606,000	0.995
792	4,867,000	0.991
424	4,121,000	0.987
114	3,022,000	0.984
185	2,496,000	0.981
201	1,800,000	0.978
977	1,293,000	0.975
644	1,022,000	0.972
663	836,000	0.969
109	423,000	0.966
514	383,000	0.963
577	200,000	0.961

a. (0.75 point)

Calculate the allocated capital for each of the lines of business using the proportional allocation method with 1-in-100 TV@R.

b. (1 point)

Calculate the allocated capital for each of the lines of business using the proportional allocation method with 1-in-100 WTV@R.

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

SPRING 2018 EXAM 7, QUESTION 21	
TOTAL POINT VALUE: 1.75	LEARNING OBJECTIVE(S): C3, C4
SAMPLE ANSWERS	
Part a: 0.75 point	
<u>Sample 1</u> 1-in-100 TV@R for personal property = average of losses to simulation #709 down to #663 since $1,000 \times 0.01 = 10$ = 3,199,600 Capital allocated: Personal auto = $10M \times 4.5M / (4.5M + 3.1996M) = 5.8445$ million Personal property = $10M \times 3.1996M / (4.5M + 3.1996M) = 4.1555$ million <u>Sample 2</u> Prop 1-100 TVAR = $(6,933,000 + \dots + 836,000) / 10 = 3,199,600$ Auto: $4.5 / (4.5 + 3.2) \times 10 = 5.84M$ Prop: $3.2 / (4.5 + 3.2) \times 10 = 4.16M$	
Part b: 1 point	
<u>Sample 1</u> WTVAR Property = $(6,933 \times 0.5 + 5,606 \times 0.4 + 4,867 \times 0.1) / 10 = \\$6.1956M$ (near \$6.2M). Auto capital = $\\$10M \times 5 / (6.2 + 5) = \\$4.46M$ Property capital = $\\$10M \times 6.2 / 11.2 = \\$5.54M$ <u>Sample 2</u> WTVaR personal prop = $(0.001 \times 4867 + 0.004 \times 5,606 + 0.005 \times 6,933) / 0.01 = 6,195,600$ Personal property alloc = $6,195,600 / (6,195,600 + 5,000,000) \times 10M = 5,533,960$ Auto = $10M - 5,533,960 = 4,466,040$ Auto gets the rest of the capital after allocating it to property.	
EXAMINER'S REPORT	
Candidates were expected to understand how to allocate a company total capital by line of business using the proportional allocation method with different risk measures (1-in-100 TV@R and 1-in-100 WTV@R here). For this, they were also expected to understand these risk measures and know how to calculate them.	
Part a	
Candidates were expected to know how to calculate 1-in-100 TV@R value for a company's line of business using a certain number of large loss simulations. They were also expected to be able to allocate the company capital proportionally by line of business using each line's TV@R.	

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

Common mistakes included:

- Using incorrect number of simulations for the property TV@R calculation.
- Using the V@R instead of the TV@R for property (or a loss value near the V@R).
- Using the WTV@R instead of the TV@R for property.

Part b

Candidates were expected to know how to calculate 1-in-100 WTV@R value for a company's line of business using a predefined transformed probability distribution for one of the lines.

Candidates were expected to understand the WTV@R measure and know which loss simulations to use in the calculation with their right probability (capped probability for the last simulation).

Candidates were also expected to be able to allocate the company capital proportionally by line of business using each line's WTV@R.

Common mistakes included:

- Using inadequate weights for the WTV@R calculation.
- Forgetting to cap the probability of the last simulation.
- Using the TV@R instead of the WTV@R for auto.
- Using the V@R instead of the WTV@R for property (or a loss value near of the V@R).
- Using the 10 worst or the 13 given loss simulations in the property WTV@R calculation.
- Weighting the loss simulations with $F^*(x)$ instead of $f^*(x)$.